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# Research on Practical Inference Acceleration of Video Generation Models Based on Spatial Token Merging

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## Abstract

With the rapid development of video generation models such as CogVideoX and Hunyuan-Video, inference efficiency has become an important practical problem. Large video diffusion Transformer models usually require long inference time and large GPU memory, which limits their use under limited computational resources. This project aims to study a practical and training-free acceleration method for video generation models based on spatial Token Merging. Instead of designing a new model or retraining a large video generation model, we plan to insert a lightweight token merging module into selected Transformer blocks of an existing open-source model. The method will utilize a bipartite matching strategy to merge visually similar tokens within each video frame while keeping the overall model structure unchanged. Through qualitative and quantitative experiments—utilizing metrics such as FVD and CLIP Score—this project will analyze the trade-off between generation quality and inference efficiency, and provide a reproducible implementation suitable for a course project.

## 1 Background and Motivation

In recent years, generative AI has made significant progress in the video domain. Architectures represented by diffusion Transformers have achieved high-quality video synthesis by modeling high-dimensional latent spaces.

However, video generation models are usually much more expensive than image generation models. A video contains many frames, and each frame contains many spatial tokens. As a result, the self-attention computation during inference becomes a major bottleneck, especially when the model needs to generate videos with higher resolution or more frames.

From the perspective of model efficiency, video data contains strong spatial redundancy. Neighboring patches in the same frame may share similar texture, color, or semantic information. Therefore, it may be unnecessary to compute full attention for every token at every layer. Inspired by Token Merging in Vision Transformers, this project plans to reduce redundant token computation during inference.

To make the project stable and feasible within the course schedule, we will focus on a conservative version of token merging. Instead of aggressively merging tokens across the whole spatiotemporal sequence, we will mainly perform frame-wise spatial token merging and keep the input and output shapes of Transformer blocks unchanged. This design reduces implementation risk and makes the method easier to compare with the original baseline.

Therefore, this project plans to conduct research in the following dimensions:

- **Identifying Spatiotemporal Redundancy:** Investigate how to use intermediate features during forward passes to identify redundant visual tokens. The main focus will be spatial redundancy within each frame, while temporal consistency will be considered in the evaluation.

- **Constructing Acceleration Mechanisms:** Implement a plug-and-play spatial Token Merging module that can be inserted into selected Transformer blocks of an existing pre-trained video generation model without retraining the whole model.
- **Systematically Analyzing Quality Trade-offs:** Analyze how different token merging ratios affect inference speed, GPU memory usage, and video quality. The project will focus on practical trade-offs rather than pursuing the maximum possible acceleration.

## 2 Related Work

The acceleration of diffusion-based generative models has become an important research direction. Existing approaches can be broadly categorized into two streams:

**Algorithmic Step Reduction.** Techniques such as Latent Consistency Models and distillation-based methods aim to reduce the number of denoising steps during inference. These methods can significantly reduce latency, but they often require additional training or fine-tuning. For large-scale video generation models, this training cost can be too high for a course project with limited computational resources.

**Architectural Redundancy Elimination.** Token Merging was originally introduced to accelerate Vision Transformers by merging redundant visual tokens during inference. This idea is attractive because it can be applied without retraining in many settings. Some related methods have explored token reduction or sparse attention in diffusion models. However, directly applying aggressive token merging to video generation may cause flickering, structure distortion, or unstable motion. Therefore, our project adopts a more conservative strategy: frame-wise spatial token merging with controlled merging ratios and selected insertion layers.

## 3 Research Objectives and Proposed Methodology

Using an open-source video generation model such as CogVideoX as the main baseline, the core proposed methodology is as follows:

**Spatial Token Merging Strategy.** We plan to introduce a Token Merging module into selected Transformer blocks of the video diffusion model. For each video frame, tokens with high feature similarity will be matched and merged before the attention computation. To avoid the  $O(N^2)$  complexity of exhaustive token comparison, the tokens within each frame are partitioned into two sets,  $\mathbb{A}$  and  $\mathbb{B}$ . We compute the cosine similarity only between these sets to identify the most similar pairs for merging using a bipartite matching strategy.

Instead of simple averaging, the merging process will utilize weighted feature aggregation based on the number of original tokens each merged token represents. After the attention operation, the merged representations will be restored or scattered back to the original token layout so that the output shape of the block remains unchanged. This makes the method easier to integrate with residual connections, positional embeddings, and the final video decoder.

**Dynamic Similarity Evaluation.** To avoid large additional computational overhead, we plan to reuse feature vectors already available inside the Transformer block to compute token similarity. We will start with small merging ratios, such as 10%, 20%, and 30%, and evaluate how the merging ratio affects both speed and quality. Early and late layers may be kept unchanged to preserve visual structure and final details.

## 4 Expected Outcomes and Deliverables

On a single NVIDIA RTX 4090 GPU or a comparable GPU environment, we expect to achieve the following:

- **Efficiency Metrics:** A moderate reduction in end-to-end inference latency, with a realistic target of around 10%–30% under selected token merging configurations. We will also measure average denoising step time and peak GPU memory usage.

- **Quality Assurance:** Maintain acceptable visual quality under conservative merging ratios. We will utilize metrics such as **FVD (Fréchet Video Distance)** to assess temporal consistency and video distribution quality, and **CLIP Score** to measure the semantic alignment between generated videos and input prompts. We will also compare generated videos with the original baseline and analyze possible artifacts such as texture loss, object distortion, or temporal inconsistency.
- **Final Deliverables:** 1) A modular PyTorch implementation of spatial Token Merging for the selected video generation baseline; 2) A technical report following the course-required NeurIPS-style format, including experimental results and trade-off analysis between merging ratio, inference speed, and visual quality; 3) A demonstration video or generated examples showing the comparison between the baseline model and the accelerated version.

## 5 Project Schedule

- **Weeks 1–2:** Run the baseline CogVideoX inference pipeline successfully, prepare test prompts, and record baseline inference time, GPU memory usage, and FVD/CLIP scores.
- **Weeks 3–4:** Implement the frame-wise spatial Token Merging module with bipartite matching, insert it into selected Transformer blocks, and verify that the modified model can generate videos correctly.
- **Weeks 5–6:** Conduct experiments with different merging ratios and insertion layers, organize experimental charts and qualitative examples, and complete the final report and presentation.

## References

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